

A Proofs

Proof of Theorem 1. Given an input parameter pair (ϵ, μ) , by the definition of (ϵ, ρ) -neighbors, for any vertex v , the number of its (ϵ, ρ) -neighbors lies in the interval $[d_{\epsilon+\rho}(v), d_{\epsilon-\rho}(v)]$. By the definition of core vertices, the core vertex set produced by the ρ -approximate SCAN with parameters (ϵ, ρ, μ) satisfies

$$(\epsilon + \rho, \mu)\text{-core} \subseteq (\epsilon, \rho, \mu)\text{-core} \subseteq (\epsilon - \rho, \mu)\text{-core},$$

where $(\epsilon, \rho, \mu)\text{-core}$ denotes the core vertex set under parameters (ϵ, μ) . According to Properties 2 and 3 of SCAN, the inclusion relationships are preserved across the corresponding clustering results. Therefore, the theorem holds.

Proof of Observation 1. Given an input parameter pair (ϵ, μ) , once the clustering result $C_{\epsilon, \mu}$ is explicitly materialized as a collection of clusters, the space cost is $O(n)$, since only vertex identifiers need to be stored and neither edges nor similarity scores are required. Moreover, clusters can be directly retrieved without performing a BFS-style traversal. Therefore, the observation holds.

Proof of Theorem 2. Given an input parameter pair (ϵ, μ) , by the definition of core vertices, a vertex v is a non-core vertex if and only if $d_\epsilon(v) < \mu$. Assume that there exists a vertex $u \in N_\epsilon[v]$ such that $d_\epsilon(u) \geq \mu$. Then u is a core vertex and $u \in N_\epsilon^+[v]$. By the definition of clusters, vertex v is assigned to the cluster containing u , and the theorem holds.

Proof of Theorem 3. After uniform discretization of the similarity range, consider an input parameter pair (ϵ, μ) . If ϵ lies exactly on a discretization boundary, the exact clustering result can be obtained by directly applying an exact SCAN algorithm. Otherwise, suppose that $\epsilon \in [(i-1)/\delta, i/\delta)$. In this case, we return the clustering result $C_{(i-1)/\delta, \mu}$, i.e., the clustering result under parameter pair $(\frac{i-1}{\delta}, \mu)$. By construction, we have

$$\epsilon - \frac{1}{\delta} < \frac{i-1}{\delta} < \epsilon + \frac{1}{\delta},$$

which implies

$$(\epsilon - \frac{1}{\delta}, \mu) \prec (\frac{i-1}{\delta}, \mu) \prec (\epsilon + \frac{1}{\delta}, \mu).$$

Therefore, by the definition of ρ -approximate SCAN and the nested property of SCAN, the returned clustering is a ρ -approximate result with $\rho = \frac{1}{\delta}$. The theorem thus holds.

Proof of Theorem 4. Given an input parameter pair (ϵ, μ) . For a vertex v and a PPT $(\epsilon', \mu') \in PPT(v)$, we have v has μ' ϵ' -neighbors. If $(\epsilon, \mu) \prec (\epsilon', \mu')$, we have $\epsilon' \geq \epsilon$ and $\mu' \geq \mu$. Therefore, v has at least $\mu' \geq \mu$ ϵ -neighbors, i.e. v is a core vertex under (ϵ, μ) . The theorem thus holds. Given an input parameter pair (ϵ, μ) , consider a vertex v and a PPT $(\epsilon', \mu') \in PPT(v)$. By definition of PPT, vertex v has at least μ' neighbors with similarity no smaller than ϵ' . If $(\epsilon, \mu) \prec (\epsilon', \mu')$, then $\epsilon' \geq \epsilon$ and $\mu' \geq \mu$. Therefore, all ϵ' -neighbors of v are also ϵ -neighbors, and v has at least $\mu' \geq \mu$ ϵ -neighbors. Hence, v is a core vertex under parameters (ϵ, μ) . The theorem holds.

Proof of Theorem 5. Consider a vertex v with degree $d(v) \geq 1$ and δ discretized similarity intervals. If $N(v) = \{u\}$, then v has at least one PPT $(\sigma(u, v), 2)$, and thus $p(v) \geq 1$. If $d(v) \leq \delta$, the maximum number of PPTs of v is $d(v)$, which occurs when all similarities $\sigma(v, u_1)$ and $\sigma(v, u_2)$ are distinct for any $u_1, u_2 \in N(v)$ with $u_1 \neq u_2$. If $d(v) > \delta$, the maximum number of PPTs of v is δ , which occurs when for each interval $i \in [1, \delta]$, there exists a

neighbor $u \in N(v)$ such that $\sigma(u, v) \in [(i-1)/\delta, i/\delta)$. Therefore, for any vertex v with $d(v) \geq 1$, we have

$$1 \leq p(v) \leq \min\{d(v), \delta\}.$$

The theorem holds.

Proof of Theorem 6. First, we prove correctness, i.e., the connectivity among all cluster slices can be correctly recovered at query time. Assume that two cluster slices $S_1(\epsilon_1, \mu_1)$ and $S_2(\epsilon_2, \mu_2)$ are connected under an input parameter pair (ϵ, μ) , where $\epsilon \leq \min\{\epsilon_1, \epsilon_2\}$ and $\mu \leq \min\{\mu_1, \mu_2\}$.

We consider the following three cases.

Case (1): S_1 and S_2 can be connected through other cluster slices. In this case, the connectivity between S_1 and S_2 can be recovered by transitively connecting them via these intermediate cluster slices using the stored connectivity.

Case (2): S_1 and S_2 cannot be connected through other cluster slices, and there exists at least one edge (u, v) between them such that $\sigma(u, v) \geq \min\{\epsilon_1, \epsilon_2\}$. In this case, (u, v) satisfies the conditions of Strategies 2–5 and is therefore stored in PPT-Index, which allows S_1 and S_2 to be directly connected.

Case (3): S_1 and S_2 cannot be connected through other cluster slices, and for all edges (u, v) between them, $\sigma(u, v) < \min\{\epsilon_1, \epsilon_2\}$. By Strategy 3, none of these edges is stored. However, since S_1 and S_2 are connected under (ϵ, μ) , there must exist an edge (u, v) such that $\sigma(u, v) \geq \epsilon$. Consequently, u and v belong to cluster slices $S_3(\sigma(u, v), \mu_3)$ and $S_4(\sigma(u, v), \mu_4)$, respectively, and the edge (u, v) is stored to connect S_3 and S_4 . Thus, S_1 and S_2 can be connected transitively through S_3 and S_4 .

In all cases, the connectivity between S_1 and S_2 can be correctly recovered. Therefore, the correctness holds.

Then, we prove that the total number of stored connections is minimal and is bounded by $|S|$, where $|S|$ denotes the total number of cluster slices.

Assume, for the sake of contradiction, that there exist two redundant edges (u_1, v_1) and (u_2, v_2) that induce the same slice-level connectivity and are both stored under Strategies 2–5. Let $u_1, u_2 \in S_1$ and $v_1, v_2 \in S_2$. If $S_1 \cap S_2 \neq \emptyset$, then neither edge should be stored according to Strategy 2 or Strategy 5. Otherwise, Strategies 3–5 ensure that at most one edge is stored for any pair of connected cluster slices based on the connectivity threshold. In both cases, the assumption that two redundant edges are stored leads to a contradiction. Therefore, the number of stored connections in PPT-Index is minimal.

In the worst case, where all cluster slices are distinct and form a connected structure, at most $|S| - 1$ connections are required to connect all slices. Hence, the total number of stored connections is bounded by $O(|S|)$.